Lecture 1 — Introduction to Econometrics and the Linear Regression Model

Econometrics L3 — Télécom Paris

Màxim Sandiumenge i Boy

Readings: Wooldridge (5th ed.) ch. 1–2

Course logistics

Course objectives

By the end of the course, you should be able to

- use the basic techniques needed to specify and estimate simple econometric models
- recognize the practical difficulties involved in modeling economic phenomena
- distinguish the assumptions and conditions required for a model's different uses: estimation, prediction or forecasting, and policy evaluation or simulation
- carry out the econometric workflow: formulate the question, choose the model and data, estimate, validate, and interpret
- estimate and interpret linear regression models with Python, diagnose common threats to credible interpretation, and state the model's domain of validity
- use introductory time-series models to analyze dynamics and produce forecasts

Course calendar: weeks 1–4

Week 1: 14 September Introduction and simple regression (3h).
Start forming project groups and looking for data.

Week 2: 21 September Multiple regression; Tutorial 1.
Bring a possible research question and data source.

Week 3: 28 September Statistical inference, including t and F tests; Tutorial 2.
Send the graded proposal to me before 28 September, 09:00.

Week 4: 5 October OLS asymptotics; combined Tutorials 3–4.
Use proposal feedback to refine the analysis.

Course calendar: weeks 5–8

Week 5: 12 October Functional form and qualitative information (90 min); heteroskedasticity and model assessment (90 min).

Week 6: 19 October Introduction to time series with a data demonstration (90 min); combined Tutorials 5–6 (90 min).

Week 7: 2 November Cross-sectional written exam (90 min); Tutorial 7: time series data lab (90 min).

Week 8: 9 November Project presentations (3h). Groups of four; 20 minutes per group.

The exam covers the cross-sectional material in Lectures 1–6.

Assessment

- **Participation and tutorial work: 15%.**
  Attempt, solve and explain exercises; written and computer work count.
- **Written exam: 45%.**
  90 minutes in week 7, covering the cross-sectional core (Lectures 1–6).
- **Project: 40%.**
  Graded proposal, short report, reproducible code and presentation.

Project: answer an economic question with data

Groups of four: nine groups for a class of 36.

1. Formulate a focused research question and explain why it matters.
2. Find suitable data; document their source, units and coverage.
3. Show motivating evidence with a clear descriptive figure or table.
4. Estimate, compare and interpret suitable regression specifications.
5. Explain uncertainty, assumptions and limits to the conclusions.

The week-3 proposal: question, data and evidence

Send me **1–2 pages before 28 September 2026, 09:00**, prepared outside class, with:

- the research question and its economic motivation;
- an accessible data source, sample, unit of observation and key variables;
- one initial descriptive figure or table, with an interpretation;
- a feasible analysis plan and one important limitation.

The proposal is **graded within the 40% project component**. We assess clarity, feasibility, evidence and coherence; a completed regression analysis is not yet required.

Project proposal approval

Proposals are reviewed for approval **in the order I receive them**.

Approval requires:

- a clear, well-motivated economic research question;
- a good fit between the question and the available data;
- meaningful preliminary descriptive analysis, with interpretation;
- sufficient ambition and a feasible tentative analysis plan.

Approval reserves the data and research question for your group. Groups proposing similar projects will need to find alternatives.

Final project submission and presentation

- **Short report: maximum 5 pages in total, including text, figures and bibliography.** Explain the question, data work, methods, results and limitations.
- **Reproducible code:** include a README and the data or access instructions.
- **Submit before the final session:** report, code and presentation slides. Include a short statement of each member's contribution.
- **Final-day presentation:** 20 minutes per group of four. Every member speaks and understands the analysis.

Use the final 15 minutes of tutorials for project questions throughout the course.

Lecture 1 roadmap

Today's roadmap

1. What is econometrics?
2. Why social data are difficult: measurement, variation, and identification
3. Data, modeling, and a working wage example
4. Three uses of econometric models
5. The simple and multiple linear regression model
6. Population projection, OLS, and the Gauss–Markov theorem
7. Interpreting regression output: *ceteris paribus*

What is econometrics?

What is econometrics?

Econometrics = statistical methods applied to economic data to

- **test economic theories** (does the theory survive confrontation with data?)
- **estimate economic relations** (by *how much* does y respond to x?)
- **evaluate policies** and **forecast**

Econometrics is set apart from mathematical statistics by the problems inherent in collecting and analyzing non-experimental economic data: identification, measurement, and interpretation.

What is econometrics?

Econometrics = statistical methods applied to economic data to

- **test economic theories** (does the theory survive confrontation with data?)
- **estimate economic relations** (by *how much* does y respond to x?)
- **evaluate policies** and **forecast**

Econometrics is set apart from mathematical statistics by the problems inherent in collecting and analyzing non-experimental economic data: identification, measurement, and interpretation. Examples of questions:

- Does one more year of education raise wages? By how much?
- Do harsher sentences reduce crime?
- Will inflation rise next quarter?

Two motivating examples

Gary Becker's economics of crime (1968)

- Crime as a rational choice: expected gains vs. expected costs (probability of arrest $\times$ punishment, foregone legal wage as opportunity cost)
- Testable implication: crime decreases with detection probability and legal-sector opportunities

Human capital theory (Becker, Mincer)

- Education is an investment: forego earnings today for higher productivity/wages tomorrow
- At this stage, think only of an unspecified relationship:

$$\text{wage} = f(\text{education}, \text{experience}, \text{training}, \ldots)$$

- We would like to estimate how these arguments causally affect wages, holding other relevant factors fixed.

Why social phenomena are difficult to analyze

- Some economic concepts do not have a unique **empirical measure**: should productivity be measured as output per worker, output per hour, or value added per worker?
- Data relevant to a particular question, population, or moment in time may be unavailable or may not have been collected for that purpose.
- Some relevant concepts are not directly observed (ability, motivation); available proxies may be incomplete or measured with error.
- We need **variation** in a variable to identify how outcomes move with it.
- Many dimensions of the environment interact, and we rarely observe or control all of them.

What must be made explicit?
How observations were selected, how variables were measured, where the useful variation comes from, and which factors the comparison is intended to hold fixed.

Economic data

Economic data: definitions, types, and sources

Definitions and variable types

- **Discrete qualitative**: employed/unemployed, occupation
- **Discrete ordered**: age, education, tenure, number of patents
- **Continuous**: temperature, height, unemployment rate
- **Continuous by approximation**: income, hours worked, GDP

Sources

- Statistical surveys
- Administrative records
- Purpose-built studies and experiments
- Digital traces, logs, and web data

From economic concepts to measured variables

An economic concept can be measured in different ways. Check how each variable is defined and constructed, and which population it covers, before interpreting it or comparing it across datasets or dates.

Main data structures

- **Cross-section:** many units observed at one point in time.
  Baseline in this course: units are randomly sampled from a population, so observations are treated as independent.
- **Time series:** one unit or system observed over time.
  Observations are generally not independent because variables are linked across dates.
- **Pooled cross-sections:** independent cross-sections at different dates
  Example: housing prices sampled before and after a tax reform.
- **Panel data:** the same units followed over time
  Example: 100 firms observed annually from 2010 to 2020.

This course focuses first on cross-sections, then introduces time series.

The data-collection process is central to causal analysis

The process used to collect data is key to producing a causal analysis.

Controlled experiment

- control the environment and balance relevant characteristics
- introduce one controlled difference between treatment and control groups
- costly and sometimes materially or ethically impossible
- may miss equilibrium effects when a treatment is generalized

Observational data

- variables of interest are not randomly assigned
- people choose education partly according to ability, family background, motivation, ...
- these same factors may also affect wages

The comparison must be justified

Why should the observed differences isolate the change whose effect we want to measure?

Modeling

Modeling: from a question to an answer

The objective is usually to answer a question about a **causal mechanism**.

1. Use available theory and institutional knowledge to identify the mechanism and the variables involved
2. Collect suitable data on those variables — and on relevant background factors
3. Specify quantitative relationships and state their assumptions
4. Estimate the parameters and quantify uncertainty
5. Check whether the assumptions needed for estimation and interpretation are credible
6. Answer the original question within the model's domain of validity

Economic model A conceptual account of behavior or mechanisms

Econometric model A quantitative, stochastic specification that can be confronted with data

Modeling the environment: what enters, what remains outside

- A social outcome is influenced by many observable and unobservable factors: institutions, aggregate conditions, preferences, information, and individual heterogeneity.
- A useful model is a **purposeful simplification**: retain the first-order effects on average, while identifying the influences left out and considering how they can be characterized or controlled.
- Included variables describe the mechanism and the environment we can control for.
- A different question or environment may require a different simplification.

What should remain explicit?
Which mechanisms are retained, which influences are left outside, and why this simplification is appropriate for the question and the available data.

Why conditioning matters: Simpson's paradox

Aggregated data

	Recovered	Rate
Treatment	20/40	50%
Control	16/40	40%

Treatment appears better.

Condition on a relevant group characteristic

Group	Assignment	Recovered	Rate
A	Treatment	18/30	60%
A	Control	7/10	70%
B	Treatment	2/10	20%
B	Control	9/30	30%

Within *both* groups, treatment appears worse.

The reversal comes from different proportions of high- and low-baseline groups: pooling mixes the treatment comparison with a **composition effect**.

Why it matters

It motivates careful thinking about confounding and ceteris paribus comparisons and, later, using multiple regression to condition on observed factors.

Experimental design: fertilizer and crop yield

Question: what is the effect of fertilizer quantity on crop yield?

- Consider plots in different locations, receiving different quantities of fertilizer.
- Yield also depends on observable weather (temperature, humidity, rainfall) and on soil quality, which is difficult to measure completely.
- If fertilizer quantity is chosen according to soil quality, the effects of fertilizer and soil quality cannot be separated by a simple comparison.

Design principle
Randomly assign fertilizer quantities across plots so that fertilizer quantity is independent of soil quality: $Q_j \perp \text{soil quality}_j$.

From the design to a statistical model

- Modeling is incomplete: some relevant factors are unobserved or deliberately omitted.
- Their combined impact is represented by a random error term.

For plot j, write yield per square meter as

$$R_j = f(\text{rain}_j, \text{temperature}_j, \text{humidity}_j, Q_j, \varepsilon_j),$$

where Q_j is fertilizer quantity and ε_j represents soil quality and other unobserved factors.

- Random assignment makes Q_j unrelated to ε_j by design, allowing differences in yield to reveal the causal effect of fertilizer.

General lesson
To interpret modeled variables causally, we must ensure—in an appropriate sense—that they are unrelated to the error term collecting unavailable, unobserved, or omitted factors.

A non-controlled sample: the return to education

Question: what is the effect of one more year of education on earnings?

- Human-capital theory links individual productivity to education, training, and professional experience.
- Education may be approximated by years of schooling, qualifications, training, ...
- Experience may involve years employed, occupation and responsibilities, tenure, mobility, career interruptions, ...
- However, apparently similar measures can conceal substantial heterogeneity (field and quality of education, job content, responsibilities, career interruptions), and useful analysis may require retrospective career information.

Measuring remuneration and the relevant environment

How should remuneration be measured?
It may include salaried or self-employed income, cash pay and bonuses, social protection and pensions, and benefits in kind.

Other forces can shape observed remuneration:

- The market value of a qualification can change with the number of similarly qualified workers and with economic conditions over time.
- Workers who change firms without an intervening unemployment spell generally have higher wages; the number or frequency of such job changes can be included as a variable.
- Gender, family status, children, and career interruptions can affect careers and the components of compensation.
- Responsibilities and career stage can also change remuneration.

An unobserved factor: talent

Talent or ability may affect both education and wages, but may be absent from the dataset.

$$W_j = f\left(\text{Exp}_j, \text{Edu}_j(\text{Talent}_j), \text{Training}_j, \text{Gender}_j, \text{Children}_j, \varepsilon_j(\text{Talent}_j)\right).$$

- A cross-sectional employment survey may contain retrospective career and socioeconomic information but no convincing measure of talent.
- Education and the unobserved component then share a common determinant.
- Consequently, the observed education–wage relationship need not identify the causal return to education.

Before estimation
Analyze how the variables introduced in the model are generated and related.

Different roles of variables in the wage example

- **Dependent variable of interest (endogenous)**: remuneration W_j, the variable to be modeled
- **Exogenous explanatory variable of interest**: professional experience X_j, which enters the mechanism
- **Endogenous explanatory variable of interest**: education X_j^*, whose generation process is linked to that of remuneration
- **Environment variables**: background conditions generated outside the modeled mechanism and not jointly determined with the variables of interest (e.g. common cross-sectional conditions or aggregate panel indicators)
- **Error term** ε_j: unmodeled environmental influences with diffuse, random effects, plus unobserved factors that may be correlated with endogenous explanatory variables

Course baseline: explanatory variables are treated as exogenous

Maintained assumption in the introductory course
We assume there is no systematic link between the explanatory variables and the error term: the explanatory variables are treated as exogenous.

- This is the baseline case used to introduce estimation and interpretation.
- It may fail in non-controlled samples when an omitted or unobserved factor also helps determine an explanatory variable.
- When such a link exists, we need a different research design or a more advanced estimation method.

Three main uses of an econometric model

1. **Learn about and test an economic theory**
   Estimate a parameter or test a restriction implied by a mechanism.
2. **Predict or forecast**
   Predict an outcome for a new unit or a future date, with uncertainty.
3. **Evaluate or simulate a policy**
   Estimate what would happen under a treatment, reform, or changed environment.

Use 1: learn about and test theory

Question: what is the value of a parameter or set of parameters?

- Build a model of an economic phenomenon and confront theoretical predictions with the available data.
- Interest may lie in the value or sign of one coefficient, or in a restriction involving several coefficients.
- **Example:** are nominal wages indexed to anticipated inflation?
- **Example:** does an aggregate production function $Y = AL^{\alpha}K^{\beta}$ exhibit decreasing returns, $\alpha + \beta < 1$?

Use 1: estimation and testing

- Estimate the parameters of interest as precisely as possible.
- Use a statistical test with sufficient power to make a well-founded decision about the theoretical restriction.
- Check whether economic theory links the parameters we want to estimate to other observable relationships, such as firms' input choices.
- Estimating these relationships together may improve precision, provided the additional relationships are correctly specified.

Use 2: predict or forecast

Questions: what value will a variable take at horizon h? What value is expected for a particular configuration of the environment?

- **Statistical or machine-learning predictor**: learn recurring patterns from historical data without requiring a causal mechanism; choose the method by how well it predicts held-out or future observations. It answers "what is likely?" while new cases resemble the data used to train and validate it.
- **Descriptive model**: summarize how an outcome varies with observed attributes in a population, then predict a comparable unobserved case. A hedonic housing-price model predicts a home's price from attributes such as location, floor area, age, and energy efficiency; the relationships need not be causal.
- **Structural model**: specify a causal economic mechanism. It aims to explain *why* the prediction changes and to study counterfactual scenarios, but requires stronger assumptions—for example, an inflation model based on input prices, labor costs, taxes, and oil prices.

Use 2: a forecast is a scenario with uncertainty

- Report both the expected or point forecast and its precision: uncertainty about the conditional mean is different from the range of possible individual outcomes.
- Choose values for the environmental variables that are plausible for the mechanism and forecast horizon.
- Those future explanatory values may not be freely chosen: they can depend on current or past values of the variable being forecast.
- Performance can deteriorate after structural change or an exceptional event.

Example
What happens to inflation if oil prices rise by 5%, holding the other scenario variables fixed?

Use 3: evaluate or simulate a policy

Question: what would an economic outcome be if a reform were implemented?

- A social experiment compares a properly constructed treatment and control group; sample composition and attrition must be controlled.
- A model can simulate a policy change not yet observed, provided the reform is described within a framework compatible with the data used to estimate the model.
- Behavior and expectations may adapt after implementation, so pre-reform relationships need not remain unchanged.

Policy questions require counterfactuals
We need an estimate of what outcomes would have been without the reform.

What an econometric model can — and cannot — do

- A model illuminates a selected aspect of an economic or social mechanism.
- It is constrained by the nature of the available data in two ways:
  - the available measurements may only partially fit the mechanism studied;
  - unobserved factors may prevent a ceteris paribus interpretation of some effects.
- It is designed to answer a limited set of questions, so its domain of validity is generally restricted.
- In general, it cannot answer questions other than those for which it was designed.

The linear regression model

What are we trying to learn from regression?

Much of applied econometrics begins with two population variables, y and x. We want to **explain y in terms of x**, or study how y varies when x changes.

Writing such a model forces us to confront three questions:

1. How do we allow factors other than x to affect y?
2. What functional relationship links y and x?
3. When does the relationship capture a *ceteris paribus* effect of x on y?

We can resolve these ambiguities by writing

$$y = \beta_0 + \beta_1 x + u. \tag{2.1}$$

This equation is assumed to hold in the population of interest, and defines the **simple linear regression model**.

Anatomy of the simple linear regression model

$$y = \beta_0 + \beta_1 x + u \tag{2.1}$$

- y: dependent, explained, response, or predicted variable; **regressand**.
- x: independent, explanatory, control, or predictor variable; **regressor**.
- u: error term or disturbance—the **unobserved factors affecting** y.
- β_0: intercept parameter; also called the **constant term**.
- β_1: **slope parameter**. Holding the factors in u fixed,

  $$\Delta u = 0 \quad \Longrightarrow \quad \Delta y = \beta_1 \Delta x.$$

  Thus β_1 is the effect of x on y, *ceteris paribus*.
- In the population model, y, x, and u are random variables; β_0 and β_1 are fixed parameters. Observation subscripts enter only when we draw a sample.

Example: soybean yield and fertilizer

Population model for a plot

$$\text{yield} = \beta_0 + \beta_1 \text{fertilizer} + u$$

y is soybean yield, x is the fertilizer quantity, and u contains land quality, rainfall, temperature, pests, farmer care, and other omitted factors.

Research question
How much does expected yield change when fertilizer increases, **holding the other plot features fixed**?

$$\Delta\text{yield} = \beta_1 \Delta\text{fertilizer} \quad \text{when } \Delta u = 0.$$

Random assignment does not make u disappear; it makes assigned fertilizer unrelated to the factors collected in u.

The hardest question: have we solved causality?

We have *defined* β_1 as the effect of x on y while holding the factors in u fixed. But this is not yet the end of the causality problem.

To estimate β_0 and β_1 reliably—and interpret β_1 ceteris paribus—we must restrict how u is related to x.

- Without such a restriction, changes in x can be mixed with changes in omitted factors.
- Because x and u are random variables, we need a restriction grounded in probability.

First: an assumption without loss of generality. Then: a substantive exogeneity assumption relating x and u.

An assumption without loss of generality: $\mathbb{E}(u) = 0$

As long as the model contains an intercept, we can always set

$$\mathbb{E}(u) = 0. \tag{2.5}$$

Suppose instead that

$$y = \alpha_0 + \beta_1 x + \varepsilon, \qquad \mathbb{E}(\varepsilon) = \mu \neq 0.$$

Define $u = \varepsilon - \mu$ and $\beta_0 = \alpha_0 + \mu$ (equivalently, $\alpha_0 = \beta_0 - \mu$). Then

$$y = \beta_0 + \beta_1 x + u, \qquad \mathbb{E}(u) = 0.$$

What this does—and does not—say

It normalizes the population average of the unobserved factors to zero and selects one intercept/error decomposition from infinitely many equivalent ones. It says **nothing** about the relationship between u and x—and precisely an assumption about that relationship is what we need next.

Why zero correlation is not enough

A natural first idea is to require x and u to be uncorrelated:

$$\mathrm{Cov}(x, u) = 0.$$

Correlation measures linear association only.

A counterintuitive feature

Let $U = X^2$, with $\mathbb{E}(X) = 0$ and $\mathbb{E}(X^3) = 0$. Then $\mathrm{Cov}(X, U) = 0$, even though U is completely determined by X.

It is therefore possible for u to be uncorrelated with x while being related to functions of x, such as x^2. This is not enough for the interpretation and statistical analysis we want. A better assumption uses $\mathbb{E}(u \mid x)$.

Mean independence and zero conditional mean

The crucial assumption: the conditional *mean* of u does not depend on the value of x:

$$\mathbb{E}(u \mid x) = \mathbb{E}(u). \tag{2.6}$$

- The average value of the unobservables is the same in every x-slice.
- This common average equals $\mathbb{E}(u)$; u is then **mean independent** of x.
- The conditional distribution may still change in variance or shape; only its mean is restricted.
- In practice, (2.6) must be justified from economic reasoning, institutional knowledge, or the research design.
- Alone it does not create causality: correct specification and variation in x are also needed.

Combining (2.6) with the normalization (2.5) gives

Zero conditional mean (ZCM)

$$\mathbb{E}(u \mid x) = 0. \tag{2.7}$$

What would zero conditional mean require?

Wages and education
Let x be years of education and let the only unobserved factor u be innate ability.

- (2.7) requires average ability to be the same at every education level.
- If more able people tend to choose more education, average ability rises with x and (2.7) fails.
- Because ability is unobserved, wage and education data alone do not allow us to verify (2.7); the research design must address it.

Soybean fertilizer experiment
If fertilizer quantities are assigned independently of land quality, rainfall, and other plot features in u, then (2.7) is plausible by design.

ZCM is a claim about the population and how x is generated.

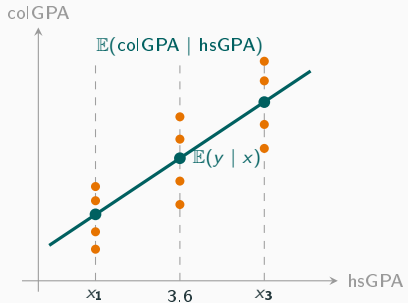
Building intuition: the slope under zero conditional mean

Taking expectations of (2.1) under (2.7):

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_1 x. \tag{2.8}$$

- The **population regression function (PRF)**, $\mathbb{E}(y \mid x)$, is a linear function of x.
- A one-unit increase in x changes the *expected value* of y by β_1.
- For any x, the mean of the conditional distribution of y is $\mathbb{E}(y \mid x)$.
- It does not say every unit satisfies $y = \beta_0 + \beta_1 x$.

GPA example:
$\mathbb{E}(\text{colGPA} \mid \text{hsGPA}) = 1.5 + 0.5\,\text{hsGPA}$.
At hsGPA $= 3.6$, expected colGPA $= 3.3$.



The dots show possible outcomes within each x-slice; no normality is assumed.

Systematic and unsystematic parts of y

Under zero conditional mean, (2.1) can be read as

$$y = \underbrace{\beta_0 + \beta_1 x}_{\mathbb{E}(y \mid x)} + \underbrace{u}_{y - \mathbb{E}(y \mid x)}.$$

Systematic part
$\beta_0 + \beta_1 x = \mathbb{E}(y \mid x)$ is the part of y explained by x: it describes how average outcomes vary across values of x.

Unsystematic part
u is the part of y not explained by x. Units with the same x can still have different outcomes because their u values differ.

ZCM centers each conditional distribution around the PRF; it does not require the unexplained component to be small.

Deriving the Ordinary Least Squares Estimates

Let $\{(x_i, y_i) : i = 1, \ldots, n\}$ be a random sample from the population. Because the data come from (2.1), each observation satisfies

$$y_i = \beta_0 + \beta_1 x_i + u_i. \tag{2.9}$$

The model implies two restrictions on the joint distribution of (x, y):

$$\mathbb{E}(y - \beta_0 - \beta_1 x) = 0. \tag{2.12}$$

$$\mathbb{E}[x(y - \beta_0 - \beta_1 x)] = 0. \tag{2.13}$$

Provided x varies in the sample, choose $(\hat{\beta}_0, \hat{\beta}_1)$ to solve the sample counterparts:

$$n^{-1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0. \tag{2.14}$$

$$n^{-1} \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0. \tag{2.15}$$

This is an example of the **method of moments**. Solving gives:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \tag{2.17}$$

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}. \tag{2.19}$$

The slope is sample covariance divided by sample variance; their common divisor cancels. Its sign therefore matches the sample covariance. These are the **ordinary least squares (OLS) estimates**.

The least-squares problem

For each observation, define the fitted value and residual:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \tag{2.20}$$

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i. \tag{2.21}$$

Residuals $\hat{u}_i$ are not the unobserved errors u_i in (2.9).

Choose $(\hat{\beta}_0, \hat{\beta}_1)$ to minimize the sum of squared residuals:

$$\min_{b_0, b_1} \sum_{i=1}^{n} (y_i - b_0 - b_1 x_i)^2$$

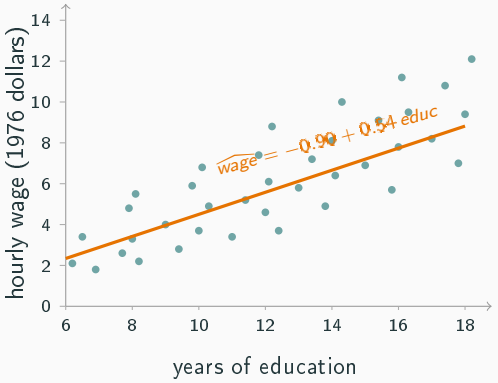
The first-order conditions are exactly (2.14)–(2.15):

$$\sum_{i=1}^{n} \hat{u}_i = 0, \qquad \sum_{i=1}^{n} x_i \hat{u}_i = 0.$$

Thus the method-of-moments estimates minimize a differentiable quadratic loss, which makes many statistical properties relatively easy to derive.

Sample regression function (SRF): $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. A new random sample generally produces a different intercept, slope, and fitted line.

Example 2.4: Wage and education



Illustrative points; fitted line from Wooldridge, Example 2.4.

$$\widehat{wage} = -0.90 + 0.54\,educ, \qquad n = 526.$$

- wage is measured in 1976 dollars per hour, and educ in years of schooling
- One more year of education raises predicted hourly wage by \$0.54
- Four more years raise it by $4(0.54) = \$2.16$ per hour
- The intercept at educ$= 0$ is an extrapolation with little economic meaning

Properties of OLS on any sample of data

Once OLS chooses $(\hat{\beta}_0, \hat{\beta}_1)$, the residuals satisfy two first-order conditions by construction:

$$\sum_{i=1}^{n} \hat{u}_i = 0, \qquad \sum_{i=1}^{n} x_i \hat{u}_i = 0.$$

The first condition directly implies

$$0 = \overline{\hat{u}} = \bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} \quad \Longrightarrow \quad \hat{y}(\bar{x}) = \bar{y}.$$

Since $y_i = \hat{y}_i + \hat{u}_i$ and $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, the two conditions also imply

$$\overline{\hat{y}} = \bar{y}, \qquad \sum_{i=1}^{n} \hat{y}_i \hat{u}_i = 0.$$

- The OLS line passes through $(\bar{x}, \bar{y})$.
- The residuals average to zero and have zero sample covariance with x.
- The fitted values and residuals also have zero sample covariance.

Properties of OLS on any sample of data: Variation decomposition

OLS decomposes each y_i into a fitted value and a residual: $y_i = \hat{y}_i + \hat{u}_i$.

Define the total sum of squares (SST), explained sum of squares (SSE), and residual sum of squares (SSR):

$$\underbrace{\mathrm{SST} \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2}_{\text{total variation}} \qquad \underbrace{\mathrm{SSE} \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}_{\text{explained variation}} \qquad \underbrace{\mathrm{SSR} \equiv \sum_{i=1}^{n} \hat{u}_i^2}_{\text{unexplained variation}}$$

Total variation in y is the sum of explained and unexplained variation: $\mathrm{SST} = \mathrm{SSE} + \mathrm{SSR}$.

Goodness-of-fit

$$R^2 \equiv \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSR}}{\mathrm{SST}}, \qquad 0 \leq R^2 \leq 1.$$

- $100R^2$: percentage of the sample variation in y that is explained by x (by the fitted line).
- $R^2 = 1$: all data points lie on the fitted line; OLS fits the sample perfectly.
- R^2 near zero: little of the variation in y_i is captured by the variation in $\hat{y}_i$.

Units of Measurement

Rescale the dependent variable

$$y_i^* = c_y y_i \quad \Longrightarrow \quad \hat{\beta}_0^* = c_y \hat{\beta}_0, \qquad \hat{\beta}_1^* = c_y \hat{\beta}_1.$$

Express wage in cents instead of dollars: $\widehat{wage}_{\text{cents}} = -90 + 54\, educ$.

Rescale the explanatory variable

$$x_i^* = c_x x_i \quad \Longrightarrow \quad \hat{\beta}_0^* = \hat{\beta}_0, \qquad \hat{\beta}_1^* = \frac{\hat{\beta}_1}{c_x}.$$

Express education in months instead of years: $\widehat{wage} = -0.90 + 0.045\, educ_{\text{months}}$.

The fitted values, residual pattern, and R^2 describe the same fit in different units. Coefficient interpretation must always state the units of both variables.

Functional forms

"Linear regression" means linear in the parameters. The variables themselves may enter as logarithms or other transformations.

Form	Dependent variable	Explanatory variable	Interpretation of β_1
Level-level	y	x	$\Delta y = \beta_1 \Delta x$
Level-log	y	$\log(x)$	$\Delta y \approx (\beta_1/100)\,\%\Delta x$
Log-level	$\log(y)$	x	$\%\Delta y \approx 100\beta_1 \Delta x$
Log-log	$\log(y)$	$\log(x)$	$\%\Delta y \approx \beta_1\,\%\Delta x$

- In a log-level model, β_1 is the semi-elasticity. A one-unit increase in x has the same approximate percentage effect, $100\beta_1\%$, at every x.
- In a log-log model, β_1 is the elasticity of y with respect to x.

Interpreting models with logarithms

Example 2.10: Log wage equation

$$\widehat{\log(wage)} = 0.584 + 0.083\, educ, \qquad n = 526, \qquad R^2 = 0.186.$$

One additional year of education is associated with an approximately 8.3% higher wage. The model imposes the same percentage effect at every initial wage level.

Example 2.11: CEO salary and firm sales

$$\widehat{\log(salary)} = 4.822 + 0.257 \log(sales), \qquad n = 209, \qquad R^2 = 0.211.$$

A 1% increase in firm sales is associated with an approximately 0.257% increase in CEO salary. The slope is an estimated elasticity.

Statistical properties of OLS: Unbiasedness

We return to the population model and study $(\hat{\beta}_0, \hat{\beta}_1)$ as estimators of the population parameters (β_0, β_1). To this end, assume:

SLR.1 Population model linear in parameters: $y = \beta_0 + \beta_1 x + u$.

SLR.2 Random sampling: $\{(x_i, y_i) : i = 1, \ldots, n\}$ is a random sample from the population.

SLR.3 Sample variation in x: the observed x_i are not all equal.

SLR.4 Zero conditional mean: $\mathbb{E}(u \mid x) = 0$.

Because the sample is random, the OLS estimators are random variables with sampling distributions determined by the population model and its parameters.

Unbiasedness under SLR.1–SLR.4

$$\mathbb{E}(\hat{\beta}_j \mid x_1, \ldots, x_n) = \beta_j \quad \Longrightarrow \quad \mathbb{E}(\hat{\beta}_j) = \beta_j, \qquad j = 0, 1.$$

The sampling distribution of $\hat{\beta}_j$ is centered at β_j.

Statistical properties of OLS: Variance

Unbiasedness locates the center of the sampling distribution. Variance tells us how far $\hat{\beta}_1$ can be expected to lie from β_1 across samples. Additionally, assume:

SLR.5 Homoskedasticity: $\mathrm{Var}(u \mid x) = \sigma^2$.

Since $\mathbb{E}(u \mid x) = 0$,

$$\sigma^2 = \mathbb{E}(u^2 \mid x) = \mathbb{E}(u^2) = \mathrm{Var}(u).$$

Equivalently, SLR.4 and SLR.5 state

$$\mathbb{E}(y \mid x) = \beta_0 + \beta_1 x, \qquad \mathrm{Var}(y \mid x) = \sigma^2.$$

Under SLR.1–SLR.5, conditional on $X = (x_1, \ldots, x_n)$,

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \mathrm{Var}(\hat{\beta}_0 \mid X) = \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2}.$$

More variation in x and less unexplained dispersion make OLS more precise.

Homoskedasticity: constant conditional variance

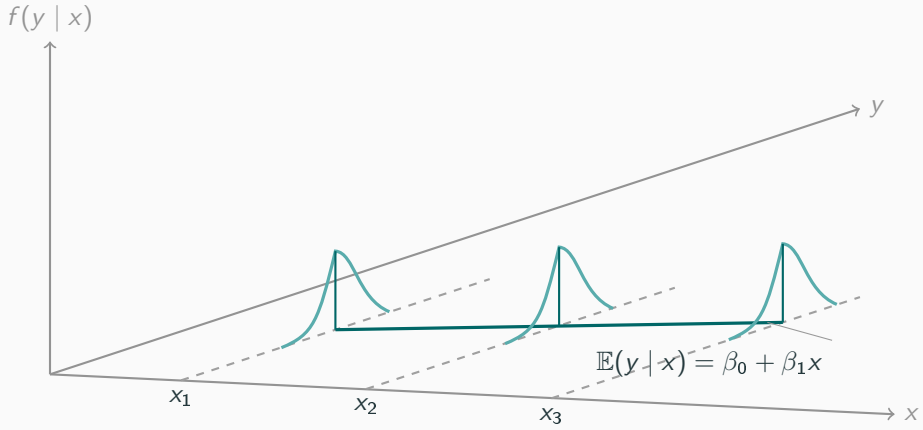


Figure 2.8. The conditional mean changes with x, while the spread of y around that mean remains constant.

Heteroskedasticity in a wage equation

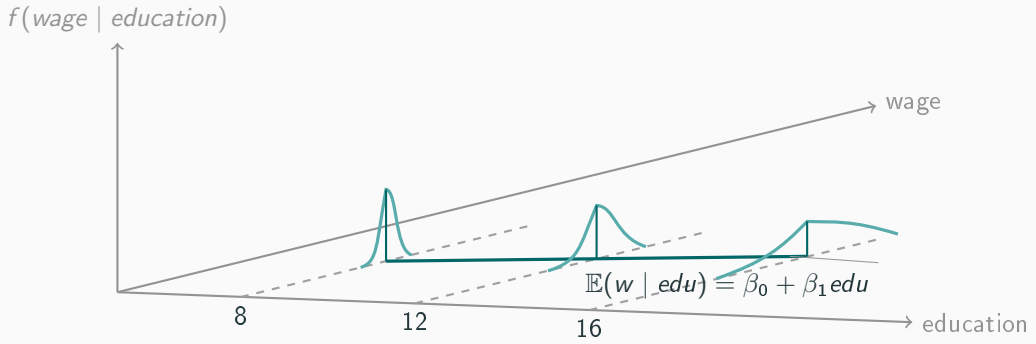


Figure 2.9. Wage dispersion may increase with education: workers with more education have a wider range of occupations and opportunities, while a wage floor compresses outcomes at low education levels.

For the remainder of this lecture, we return to homoskedasticity.

Estimating the error variance

The population errors are unobserved, so we estimate their variance from the OLS residuals.

- Population equation, with unknown parameters and an unobserved error:

$$y_i = \beta_0 + \beta_1 x_i + u_i. \tag{2.48}$$

- Sample residual, computed using the OLS estimates:

$$\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i. \tag{2.32}$$

- Using both equations:

$$\hat{u}_i = u_i - (\hat{\beta}_0 - \beta_0) - (\hat{\beta}_1 - \beta_1) x_i. \tag{2.59}$$

Since $\sigma^2 = \mathbb{E}(u^2)$, an unbiased estimator of σ^2 would be $n^{-1} \sum_{i=1}^{n} u_i^2$, but we do not observe u_i.

Using the estimates of u_i, an unbiased estimator of σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{u}_i^2 = \frac{\mathrm{SSR}}{n-2}.$$

Standard errors of the OLS estimates

Substituting $\hat{\sigma}^2$ into the variance formulas gives unbiased estimators of $\mathrm{Var}(\hat{\beta}_0 \mid X)$ and $\mathrm{Var}(\hat{\beta}_1 \mid X)$.

A natural estimator of σ is the **standard error of the regression**, $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$. It is not unbiased, but it is consistent. We use $\hat{\sigma}$ to estimate the standard deviations of the OLS estimators:

$$\mathrm{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum_i (x_i - \bar{x})^2}}, \qquad \mathrm{se}(\hat{\beta}_0) = \hat{\sigma}\sqrt{\frac{\sum_i x_i^2}{n\sum_i (x_i - \bar{x})^2}}.$$

The standard error of an estimate gives us an idea of how precise the estimator is.

Summary

- Econometrics uses data and economic reasoning to quantify relationships and assess theories or policies.
- The simple linear regression model separates the systematic component of y from unobserved factors in u.
- Zero conditional mean, $\mathbb{E}(u \mid x) = 0$, permits a ceteris paribus interpretation of the slope, but it must be justified in each application.
- OLS chooses the fitted line by minimizing the sum of squared residuals and satisfies exact sample identities.
- Under SLR.1–SLR.4, OLS is unbiased. Homoskedasticity supplies the simple variance formulas under SLR.5.
- Estimated standard errors summarize the sampling precision of the OLS estimates.

Next lecture: multiple regression and ceteris paribus comparisons with several explanatory variables.